Curriculum Vitae

PERSONAL INFORMATION



First name: Muhammad Asad, **Family name:** Lal
Email: asad@njust.edu.cn
WeChat ID: iamAsadLal
Research Profile: asadlal.tech
Date of birth: 26 January 1998
Nationality: Pakistani

AREAS OF RESEARCH

Metaheuristic and Evolutionary Optimization; Artificial Intelligence; Medical Image Analysis; Engineering Design Optimization; Machine Learning; Deep Learning

EDUCATION

08/2024 – 04/2027 Master (Computer Science and Technology)
School of Computer Science and Engineering,
Nanjing University of Science and Technology, China

08/2012 – 01/2015 Bachelor (Computer Science)
Department of Computer Science,
Government College University Faisalabad, Pakistan

PUBLICATIONS *Google Scholar*

1. **Lal, M.A.**, Kashif, M. & Huang, B. Father-inspired metaheuristic: a hybrid framework with neighborhood repulsion and elite learning for continuous optimization. ***Evol. Intel.* 19, 111 (2026).**
2. **Lal, M.A.**, Kashif, M., Lv, J. & Huang, B. AFRC-SHADE: Augmented Forman-Ricci Curvature-Guided SHADE for Multi-Level Thresholding in Fundus Image Segmentation. ***Swarm and Evolutionary Computation. Submitted* (2026).**

RESEARCH WORK

08/2024 – 04/2027 Development of Metaheuristic Algorithms
- Developed metaheuristic algorithms for continuous optimization.
- Evaluated performance using benchmark functions and statistical analysis.

08/2024 – 04/2027 Medical Image Analysis
- Applied metaheuristic optimization to medical image segmentation.
- Evaluated segmentation using quality metrics: PSNR, SSIM, and FSIM.

PROFESSIONAL EXPERIENCE

10/2023 – 03/2024 Android Development, Internship
Idealogic Software House, Punjab, Pakistan

UNDERGRADUATE PROJECTS

08/2019 – 08/2023 Heart Disease Prediction (Final Year Project)
- Developed an Android app for heart disease prediction.
- Used Logistic Regression for health classification.

02/2023 – 05/2023 Deep Learning-based Face Recognition (Semester Project)
- Developed a CNN-based face recognition model.
- Performed face recognition using image data.

FELLOWSHIPS AND AWARDS

08/2024	Fully Funded High Achievers (Lanshan) Scholarship Nanjing University of Science and Technology, China
08/2023	Outstanding Academic Performance Award Government College University Faisalabad, Pakistan

CERTIFICATIONS

06/2026	One Million Prompters Dubai Future Foundation
12/2024	Chinese Writing Competition School of International Education, Nanjing University of Science and Technology
04/2024	Manage Sensitive Data and Security Policies within GitHub Microsoft
11/2023	Machine Learning with Python International Business Machines (IBM)
11/2023	Deep Learning with TensorFlow International Business Machines (IBM)
10/2017	Computer Application and Office Professional Government of Punjab Vocational Training Council, Pakistan

TECHNICAL SKILLS

- Python
- C/C++
- LaTex
- MATLAB
- VS Code
- GitHub

LANGUAGE

- English – Fluent
- Chinese – Basic
- Urdu – Native
- Arabic – Basic

EXTRA-CURRICULAR ACTIVITIES

03/2026	Xuanwu Youth Development Activity Xuanwu District Government, Nanjing, China
08/2025	International Student Registration Volunteer Nanjing University of Science and Technology, China
06/2023	Basic Community Safety Training Punjab Emergency Service Rescue 1122, Pakistan
05/2022	BSAFE Security Awareness Training United Nations Department of Safety and Security

REFERENCE

Prof. Bo Huang
School of Computer Science and Engineering,
Nanjing University of Science and Technology, China
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